| Team Number : | apmcm2201058 |
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| Problem Chosen : | D |

# Topology and fractal optimization of heat transfer fins

## Abstract


Incorporating fins into heat storage tank is an effective approach to overcome the low thermal conductivity of phase change materials (PCMs) and enhance the heat transfer efficiency. The optimal configuration and distribution of fins are numerically obtained by topology and fractal optimization.

For question 1, a **two-dimensional finite element model** (FEM) of **computational fluid dynamics** (CFD) is established to investigate the heat transfer process. **The enthalpy-porosity method** is used to describe the phase change of PCM. The phase change starts around the rectangular fins and subsequently extends to the outer tank wall, causing the PCM to vary from the solid to liquid phase. Considering the PCM temperature rising from 293K to 333K, the initial phase change time ($t_1$) at the $\theta$ of 72°, 60°, 45°, 36°, 30° and 24° are **47.6min, 39.2min, 31.4min, 25.2min, 20.7min, 17.2min,** and the complete phase change time ($t_2$) are **158.5min, 142.9min, 123.4min, 106.2min, 92.4min, and 81.9min**, respectively. The $t_1$, liquid fraction ($f_m$), and enhancement ratio ($E_R$) are adopted as indicators to evaluate thermal behaviors. **The optimal $\theta$ is 24°** with minimum $t_1$ of 17.2min and maximum $E_R$ of 12.3%.

For question 2, the effect of $\theta$, fin length ($L$), and width ($W$) of triangular fins are systematically investigated. The results show that $\theta$=24° promotes **63.83%, 56.34%, 46.73%, 46.73%, and 31.71%** heating rate compared with $\theta$=72°, 60°, 45°, 36°, 30°. $L$=0.024m enhances **42.17%, 70.04%, and 79.52%** heating rate than $L$=0.006m, 0.012m, and 0.018m. $W$=0.01m increases by **7.60%, 22.55%, and 35.91%** heating rate than $W$=0.008m, 0.006m, and 0.004m. **The priority order of fin parameters is $L > \theta > W$**. The heating efficiency of rectangular fins at $\theta$=72°, 45°, 30°, and 24° can increase by **11.95%, 14.99%, 15.43%, and 15.62%** compared with that of triangular fans.

For question 3, two models are proposed to achieve the optimal fin distribution: topology optimization and fractal optimization. For **topology optimization**, the **variable density method** is used and **image reconstruction** is performed to achieve maximum average temperature. For **fractal optimization**, optimal parent-children fractal trees are generated by **Murray's law** and growth ratio. The results show that topology optimization can promote **18.84% and 28.17%** heating rate compared with rectangular fins and triangular fins, respectively, and fractal optimization can promote **14.01% and 24.25%**.

For question 4, a recommendation letter is written to fin company with suggestions on design optimization.


**Keywords:** heat transfer fins, CFD, topology optimization, fractal geometry.

# Contents

# 1. Introduction

## 1.1 Background

A double-wall tank with a phase change material (PCM) is commonly used for energy storage. Heated transfer liquid flows into the tank and loses energy to the PCM. The PCM absorbs heat by changing phase from solid to liquid, and releases heat from liquid to solid. Fins are embedded in the PCM between the tank walls to accelerate the heat transfer process. Using fins decreases the melting time and increases the rate of energy storage significantly. The optimal fin distribution is required for the promotion of heat transfer process.

## 1.2 Question restatement

- ✧ **Problem 1:** A cross-section with rectangular fins evenly distributed is given. Propose a model to simulate the heat transfer process in the tank. Study the effect of fin interval angle on heat transfer and calculate the transfer time when the temperature of PCM changes from 293K to 333K.
- ✧ **Problem 2:** Based on the new cross-section with triangular fins, investigate the effect of fin size on heat transfer. Compare and analyze the results with that of the rectangular fins.
- ✧ **Problem 3:** Propose a mathematical model to obtain the optimal fin distribution.
- ✧ **Problem 4:** Write a recommendation letter to then fin company for the optimal fin design.

# 2. Model hypothesis

1) The physical properties of the PCM are homogenous, isotropic, and temperature independent (except the density in the liquid state).

2) The phase change process of PCM is regarded as laminar, unsteady, and incompressible.

3) Natural convection of liquid PCM conforms to the Boussinesq assumption.

4) The liquid fraction is assumed to change linearly with temperature.

5) The effect of viscous dissipation is ignored during the simulation.

6) Heat loss to the environment is neglected

# 3. Symbol description

| Symbol | Meaning | Unit |
|---|---|---|
| $L$ | Fin length | m |
| $W$ | Fin width | m |
| $\theta$ | Interval angle | ° |

| | | |
|---|---|---|
| $T$ | Temperature | K |
| $t_1$ | Initial phase change time | min |
| $t_2$ | Complete phase change time | min |
| $R_1$ | Radius of the inner wall | m |
| $R_2$ | Radius of the outer wall | m |
| $\rho$ | Density | $kg/m^3$ |
| $f_m$ | Liquid fraction | / |
| $\lambda$ | thermal conductivity | W/(m·K) |
| $E_R$ | Enhancement ratio | / |
| $\eta$ | Penalty factor | / |
| $\alpha$ | Fractal length ratio | / |

# 4. Modeling and solution of Problem 1

## 4.1 Phase change model

A two-dimensional finite element model (FEM) was established to investigate the thermal and transient responses of the heat transfer process. The phase change of the PCM was described by the enthalpy-porosity method. ANSYS Fluent Modular based on Computational Fluid Dynamics (CFD) was used to simulate the heat transfer process.

### 4.1.1 Governing equations

The heat storage tank and cross section are depicted in Fig. 1. The interior domain $\Omega_1$ is regarded as the heat source with a temperature of 373K, and the exterior domain $\Omega_3$ is adiabatic. The middle domain $\Omega_2$ is filled with the PCM.

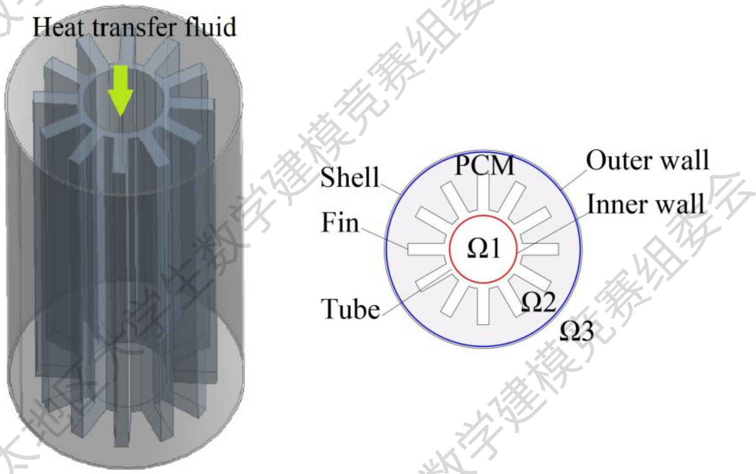



Fig. 1. Diagram of heat storage tank and cross-section

The enthalpy-porosity method is applied to describe the phase change process in the heat storage tank[1]. Governing equations of continuity, momentum, and energy equations predict the flow, heat transfer, and PCM melting, respectively[2].

**1) Continuity equation**

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0 \tag{1}$$

**2) Momentum equations**

$$\frac{\partial(\rho_{PCM}u)}{\partial t}+u\frac{\partial(\rho_{PCM}u)}{\partial x}+v\frac{\partial(\rho_{PCM}u)}{\partial y}=\frac{\partial}{\partial x}\left(\mu_{PCM}\frac{\partial u}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu_{PCM}\frac{\partial u}{\partial y}\right)-\frac{\partial p}{\partial x}+Su \tag{2}$$

$$\frac{\partial(\rho_{PCM}v)}{\partial t}+u\frac{\partial(\rho_{PCM}v)}{\partial x}+v\frac{\partial(\rho_{PCM}v)}{\partial y}=\frac{\partial}{\partial x}\left(\mu_{PCM}\frac{\partial v}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu_{PCM}\frac{\partial v}{\partial y}\right)-\frac{\partial p}{\partial y}+Sv \tag{3}$$

**3) Energy equations**

For PCM:

$$\rho_{PCM}c_{PCM}\frac{\partial T}{\partial t}+\rho_{PCM}c_{PCM}\left(u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}\right)=\frac{\partial}{\partial x}\left(\lambda_{PCM}\frac{\partial T}{\partial x}\right)+\frac{\partial}{\partial y}\left(\lambda_{PCM}\frac{\partial T}{\partial y}\right)+Se \tag{4}$$

For fin:

$$\rho_{\text{fin}} c_{\text{fin}} \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(\lambda_{\text{fin}} \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(\lambda_{\text{fin}} \frac{\partial T}{\partial y}\right) \tag{5}$$

where $T$ is the temperature, $p$ is the pressure, $u$ and $v$ are the velocities in the x and y direction, $c$ is the specific heat, $\lambda$ is the thermal conductivity, $\rho$ is the density, $Su$, $Sv$, and $Se$ are the source terms of the enthalpy method can be expressed by:

$$Su = \frac{(1-f_m)^2}{f_m^3 + \xi} u C_{mush} \tag{6}$$

$$Sv = \frac{(1-f_m)^2}{f_m^3 + \xi} v C_{mush} + \rho_{\text{PCM}} g \delta (T_{\text{PCM}} - T_{m2}) \tag{7}$$

$$Se = \rho_{\text{PCM}} L_{\text{PCM}} \frac{\partial f_m}{\partial t} \tag{8}$$

where $\delta$ is the thermal expansion coefficient, $\mu$ is the dynamic viscosity of the PCM, $f_m$ is the liquid fraction, $g$ is the acceleration of gravity, $C_{mush}$ is the mushy zone constant $C_{mush}$=$10^7$, and $\xi$ us the numerical coefficient $\xi$=$10^{-3}$ [3].

**4) Enthalpy-porosity method**

The enthalpy-porosity method uses liquid fraction to describe the state of the PCM, and assumes that the liquid phase ratio equals the porosity. The mathematical equations can be expressed as[4]:

$$f_m = \begin{cases} 0 & T \le T_{m1} \\ \dfrac{T - T_{m1}}{T_{m2} - T_{m1}} & T_{m1} < T < T_{m2} \\ 1 & T \ge T_{m2} \end{cases} \tag{9}$$

where $T_{m1}$ and $T_{m2}$ are solidus temperature and liquidus temperature, respectively.

The energy equation of this method is:

$$\frac{\partial}{\partial t}(\rho_{\text{PCM}} H) + \frac{\partial}{\partial x}(\rho_{\text{PCM}} H v_x) + \frac{\partial}{\partial y}(\rho_{\text{PCM}} H v_x) = \frac{\partial}{\partial x}(\lambda \Delta T_{\text{PCM}}) + \frac{\partial}{\partial y}(\lambda \Delta T_{\text{PCM}}) \tag{10}$$

$$\frac{\partial}{\partial t}(\rho_{\text{PCM}} H) + \frac{\partial}{\partial x}(\rho_{\text{PCM}} H v_y) + \frac{\partial}{\partial y}(\rho_{\text{PCM}} H v_y) = \frac{\partial}{\partial x}(\lambda \Delta T_{\text{PCM}}) + \frac{\partial}{\partial y}(\lambda \Delta T_{\text{PCM}}) \tag{11}$$

where $v_x$ and $v_y$ are the velocities in the x and y direction, and $H$ is the enthalpy which is calculated by the sum of the sensible enthalpy $h$ and latent enthalpy $\Delta H$:

$$H = h + \Delta H \tag{12}$$

$$h = h_{ref} + \int_{T_{ref}}^{T} c\,dT \tag{13}$$

$$\Delta H = f_m L \tag{14}$$

where $h_{ref}$ is the reference enthalpy, $T_{ref}$ is the reference temperature, and $L$ is the fin length.

#### 4.1.2 Initial and boundary conditions

**1) Initial condition**

$$T_{\mathrm{PCM}}(x,y,0)=T_{\mathrm{fin}}(x,y,0)=T_0, u=v=0, R_1^2\le x^2+y^2\le R_2^2 \tag{15}$$

**2) Boundary conditions**

$$T_{\mathrm{PCM}}(x,y,t)=T_{\mathrm{fin}}(x,y,t)=T_w, u=v=0, x^2+y^2=R_1^2 \tag{16}$$

$$\frac{\partial T(x,y,t)}{\partial n}=0, u=v=0, x^2+y^2=R_2^2 \tag{17}$$

where $T_0$ is the initial temperature, $T_w$ is the temperature of heat source, $R_1$ is the radius of the inner wall, and $R_2$ is the radius of the outer wall.

#### 4.1.3 Numerical procedure

**1) Geometry and material**

Eight fin structures with corresponding interval angles were systematically studied by ANSYS Fluent. Fig. 2 depicts the cross-section under various interval angles $\theta$. The length of rectangular fins ($L$) was 0.018m, and the width ($W$) was 0.006m. All rectangular fins are evenly distributed in the cross section. The interval angle $\theta$ changes from 180° to 30 ° accompanying fin number from 2 to 12.

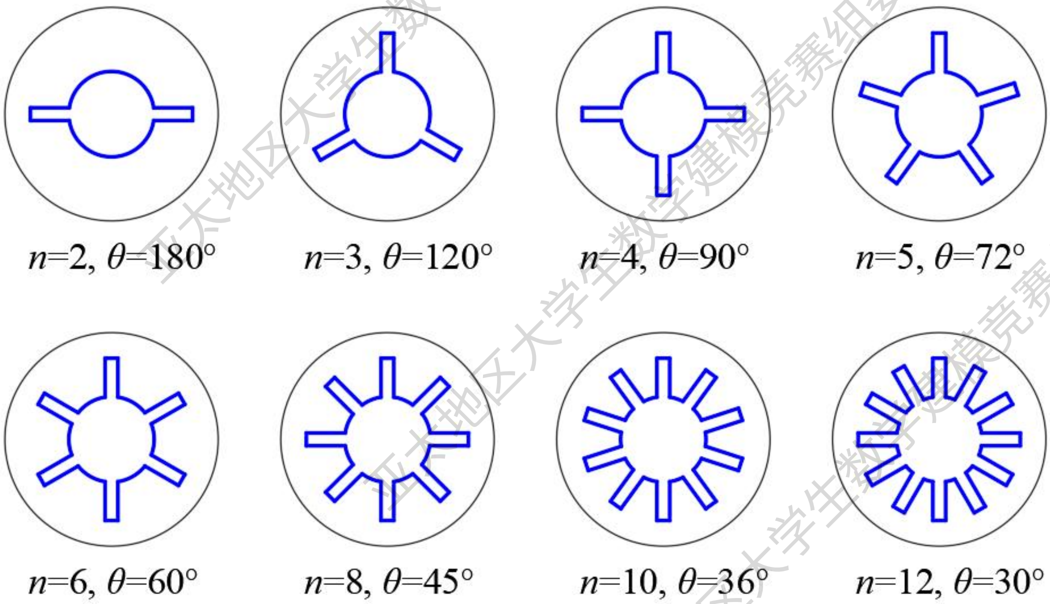



Fig. 2. Cross-section under various interval angles $\theta$

PCM can absorb or release heat energy in the process of changing from solid to liquid or from liquid to solid. The shell-and-fin tank structure (marked as Fin) and PCM domain (marked as PCM) were established in the simulation. Based on the physical properties of PCM and fin and common constants, the material parameters used in the numerical simulation are shown in Table 1.

Table 1 Material property parameters

| Symbol | Property | Unit | PCM | Fin (Quartz) |
|---|---|---|---|---|
| $\rho$ | Density | kg/m³ | 780 | 2650 |
| $\lambda$ | Thermal conductivity | W/(m·K) | 0.15 | 214 |
| $c$ | Specific heat | J/(Kg·K) | 2000 | - |
| $T_{m1}$ | Solidus temperature | K | 273.15 | - |
| $T_{m2}$ | Liquidus temperature | K | 333 | - |
| $\delta$ | Thermal expansion coefficient | $K^{-1}$ | 0.0002 | - |
| $\mu$ | Dynamic viscosity | Kg/(m·s) | / | - |

**2) Mesh size and time step**

To validate the finite element model, the independence of the grid and time step was examined. The temperature evolution was compared to obtain the accurate and efficient grid, and the phase change time was compared for the optimal time step.

✧ Mesh size

The mesh size was adjusted from 4mm (Grid1) to 1mm (Grid4) to obtain the accurate and efficient grid. Fig. 3 Shows the meshing conditions under different mesh sizes. Fig. 4(a) presents the temperature of the PCM for four grid conditions. With grid size from Grid3 to Grid4, the temperature evolutions were almost identical. Considering the computational efficiency, the mesh size in this study is 2mm.

✧ Time step

Fig. 4(b) shows that the result of phase change time is stable at the time step of 1s or 2s. Therefore, the time step used in the simulation was selected as 2s.

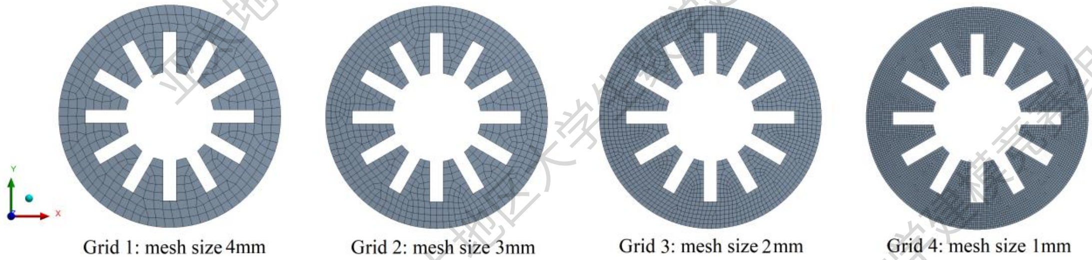



Fig. 3. Meshing conditions under different mesh sizes

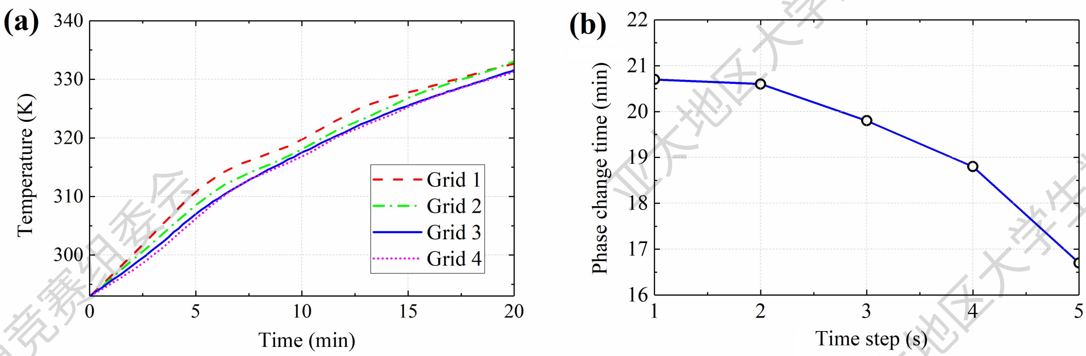



Fig. 4. Validation of the numerical model: (a) grid test, (b) time step test

### 4.2 Heat transfer process

#### 4.2.1 Temperature distribution

Fig. 5 shows the temperature distribution considering the influence of the interval angle of the fin structure. It can be seen that the phase change first starts in the area around the fins, because of the heat transfer from the fins to the PCM. As time grows, the phase change extends to the outer shell through fins, causing the PCM within the tank completely melts to liquid phase.

The addition of the fin significantly accelerates the phase change process of PCM in the system. This is because the gap between fins becomes smaller and the internal PCM can quickly absorb the heat to melt under the action of the heat conduction mechanism.

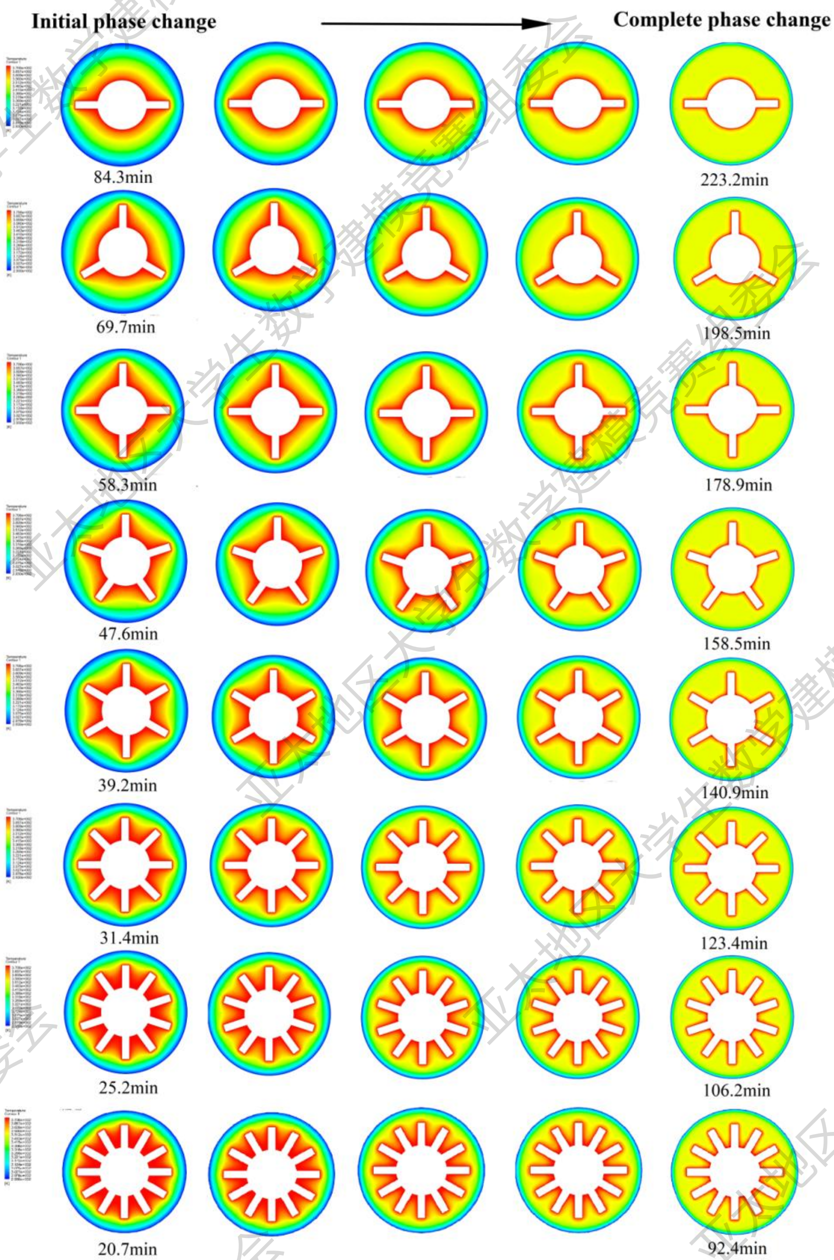



Fig. 5. Temperature distribution under different $\theta$

### 4.2.2 Phase change time

The time of initial phase change and complete phase change for PCM melting were both studied.

- ✧ **Time of initial phase change** $t_1$: it means the start time of the PCM rising from room temperature (293 K) to the phase change temperature (333K).
- ✧ **Time of complete phase change** $t_2$: it refers to the final time when the PCM completely melts to the liquid phase.

Fig. 6 shows the variation of phase change time under different interval angles. The initial and complete phase change time decreases with the decrease of the interval angle ($\theta$) or the increase of the fin number ($n$). When $\theta$ decreases from 60°, 45°, 36°, to 30°, the initial phase change time is 39.2min, 31.4min, 25.2min, and 20.7min, respectively.

However, as the decrease of $\theta$, fin efficiency weakens slightly. This is due to the shrinkage of the spacing in the middle of the tank, resulting in a slight reduction of the volume of the PCM. Therefore, the optimal interval angle should be investigated to get the most efficient heat transfer configuration.

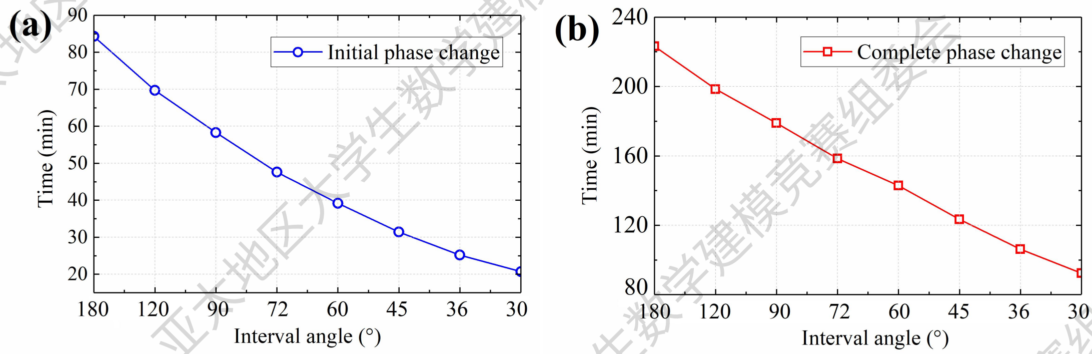



Fig. 6. Phase change time in different cases: (a) initial phase change time, (b) complete phase change time

Table 2. Phase change time

| Fin number | Interval angle (°) | Initial phase change time (min) | Complete phase change (min) |
|---|---|---|---|
| 2 | 180 | 84.3 | 92.4 |
| 3 | 120 | 69.7 | 106.2 |
| 4 | 90 | 58.3 | 123.4 |
| 5 | 72 | 47.6 | 142.9 |
| 6 | 60 | 39.2 | 158.5 |
| 8 | 45 | 31.4 | 178.9 |
| 10 | 36 | 25.2 | 198.5 |
| 12 | 40 | **20.7** | **223.2** |

### 4.2.3 Liquid fraction

Fig. 7 exhibits the variation of the liquid fraction over time under different fin numbers. It can be seen that the melting rate of the PCM increases with the increase of $n$. However, with the increase of $n$, or in other words, the decrease of $\theta$, the growth rate of melting starts to slow down to a certain critical value. The reason for this phenomenon will be explained in '4.4.2 Optimal interval angle'. Therefore, the $\theta$ of the fin should match the overall arrangement of the fins to reach optimal thermal efficiency.

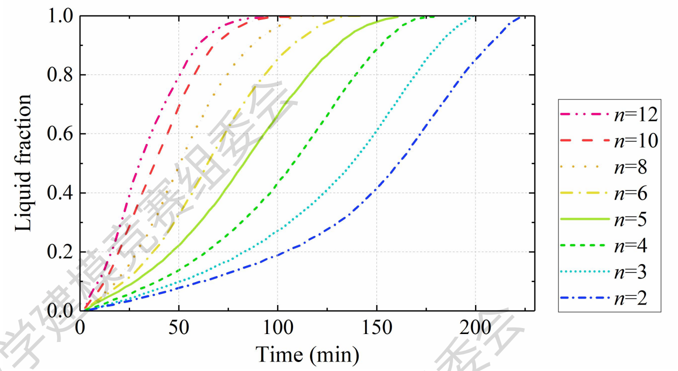



Fig. 7. Liquid fraction in different cases

## 4.3 Optimization of the interval angle

### 1) Enhancement ratio

To explore the optimal interval angle, the numerical simulation was expanded to $n$=20 to further the effect of $n$ on the heat transfer. The enhancement ratio $E_R$ was proposed to evaluate the influence of $n$ on heat transfer rate, which can be defined as:

$$E_R = \frac{f_m(n) - f_m(\text{no fins})}{f_m(\text{no fins})} \times 100\% \tag{18}$$

where $f_m(n)$ is the liquid fraction of the PCM under the $n$ fin number, and $l(\text{no fins})$ is the liquid fraction without fins added.

Fig. 8(a) shows the enhancement effects of different $n$ on the PCM melting rate. With the growth of $n$, the $E_R$ increases significantly, which indicates that larger $n$ shows a greater advantage in the immediate stage after heating. **The $E_R$ of 15 fins ($\theta$=24°) reaches the maximum value of 12.3%**, and $E_R$ of 14 fins and 16 fins are relatively low at 12.1% and 11.8, respectively.

However, with the further increase of $n$, especially at $n$ is greater than 15, the $E_R$ gradually decreases and excessive fins limit the heat transfer, causing the PCM at the wall surface to barely melt. This is because smaller spacing is left between adjacent fins and the flow resistance is larger, which weakens the amplitude of fluid movement[5].

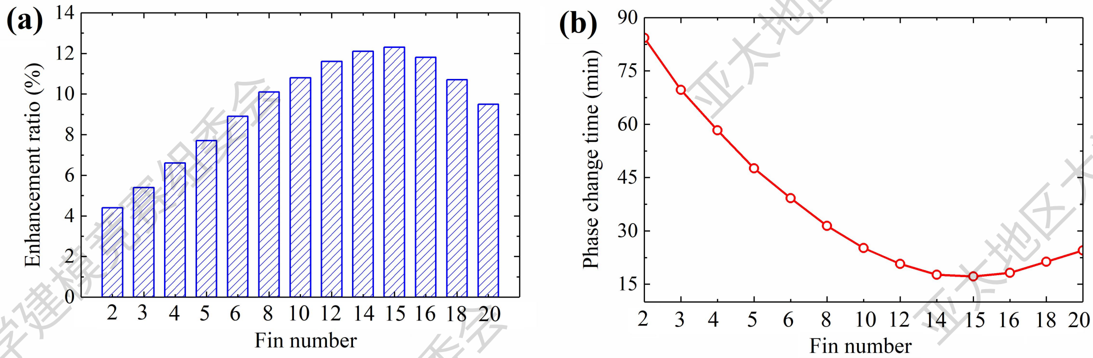



Fig. 8. Comparison of different $\theta$: (a) enhancement ratio, (b) initial phase change time

**3) Phase change time**

The initial phase change times under different $n$ were calculated correspondingly. Fig. 8(b) shows the initial phase change time of different $n$. It can still be seen that the **15-fin structure ($\theta$=24°) has the best capacity for heating with a minimum melting time of 17.2 min**.

# 5. Modeling and solution of Problem 2

## 5.1 Effect of the interval angle

Same as rectangular fins, eight interval angles of triangular fins were systematically studied in this section. Fig. 9 depicts the cross-section under various $\theta$. The $L$ was 0.018m and the $W$ was 0.006m. The $\theta$ changes from 90° to 24 ° accompanying the fin number from 4 to 15.

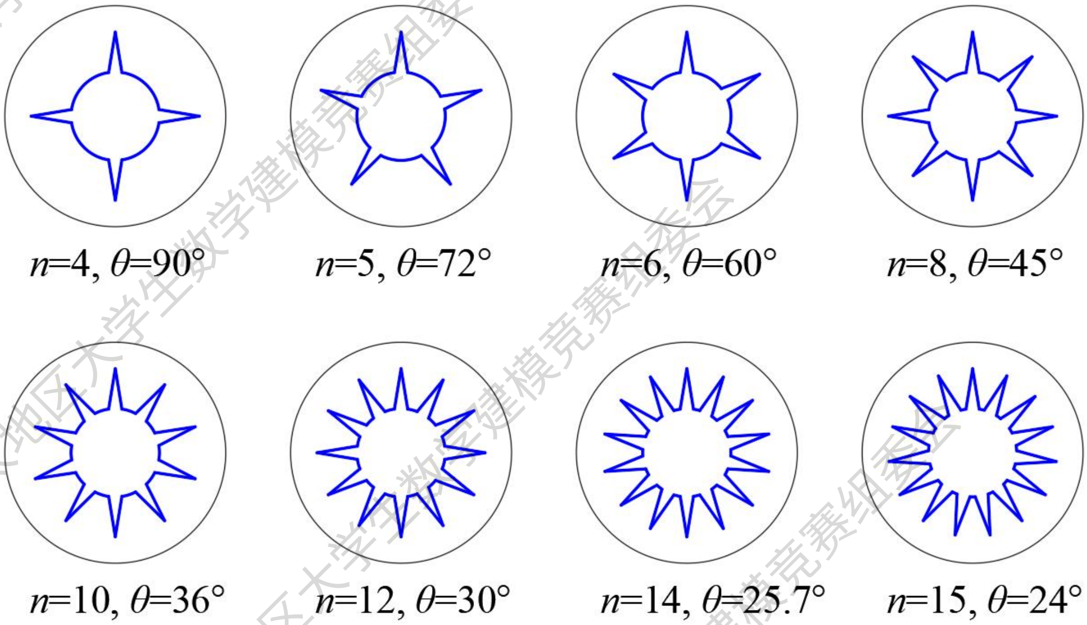



Fig. 9. Cross-section under various $\theta$

**1) Influence on heat transfer**

The performance of the heat storage was analyzed by varying the $\theta$ while keeping other parameters constant. Fig. 10 shows the temperature distribution under various $\theta$ of triangular fins. The melting time decreases by about 44.1% when the $\theta$ changes from 90° ($n$=4) to 45° ($n$=8), and significantly decreases by about 70.2% when $\theta$ equals 24° ($n$=15). This is due to the enhancement of natural convection and reduction of the thermal resistance, which are caused by the larger heat transfer surface area for more fins.

The more fins set, the faster the heating process in the system. However, the change rate of phase change time slightly decreases when the $\theta$ larger than 36° ($n$=10), especially for 26° ($n$=14) and 24° ($n$=15). This is due to the fact that the presence of fins interferes with natural convection. Therefore, there is a limit to the interval angle $\theta$ and fin number $n$.

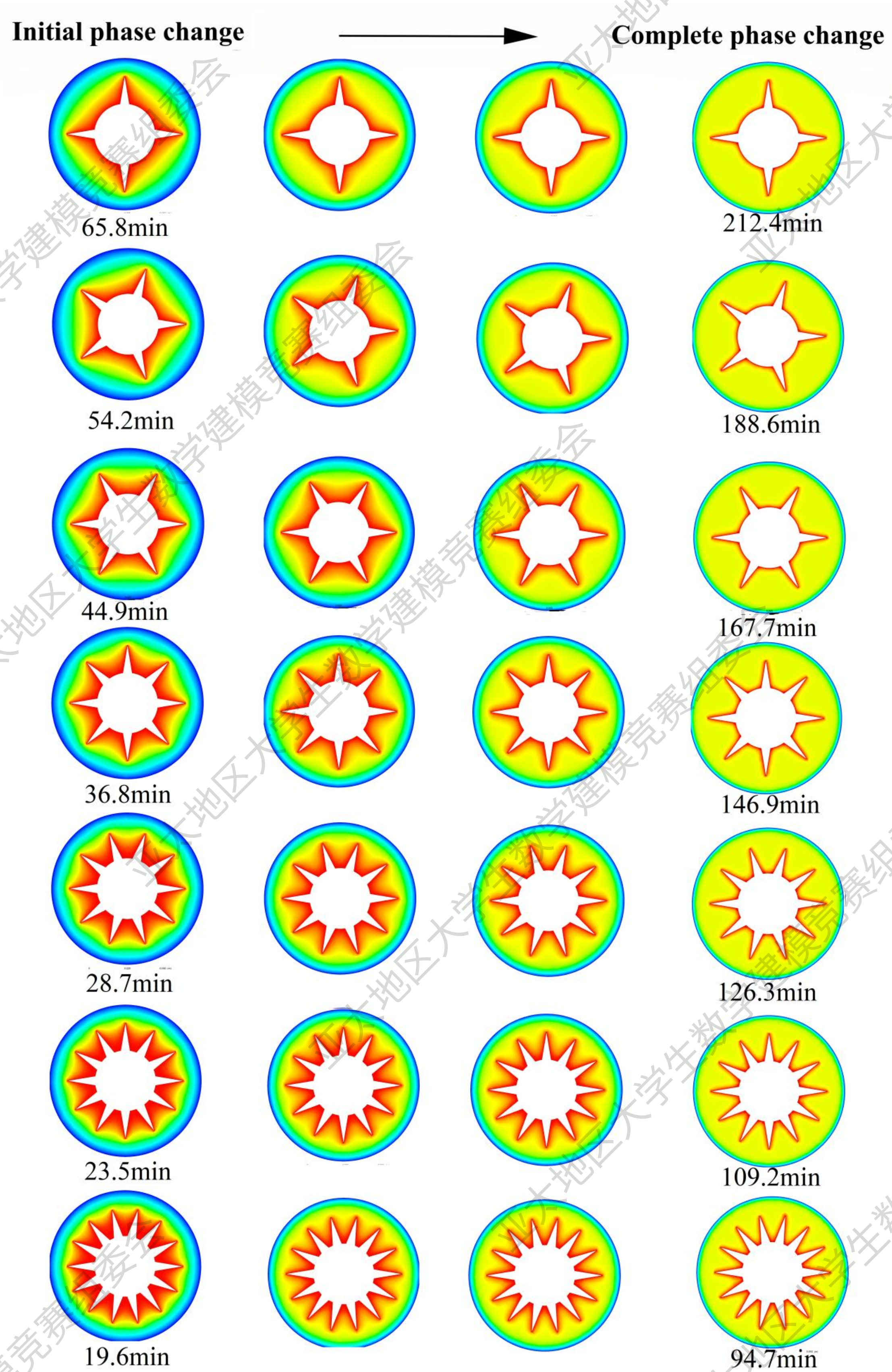



Fig. 10. Temperature distribution under different $\theta$

### 2) Comparison with rectangular fins

Fig. 11(a) shows the comparison of phase change time between rectangular fins and triangular fins for various $\theta$. The heating efficiency **gradually improves** as the fin design evolves from triangular to rectangular. $\theta$=24° promote **63.83%, 56.34%, 46.73%, 46.73%, and 31.71%** heating rate compared with $\theta$=72°, 60°, 45°, 36°, 30°. It should be pointed out that the time difference between both fins becomes smaller as the $\theta$ decreases.

Fig. 11(b) shows the liquid fraction curves for different $\theta$ during melting. The rectangular fins are more efficient than triangular fins by reducing the total melting time. Taking the $\theta$ of 72°, 45°, 30°, and 24° for examples, the heating efficiency of rectangular fins can improve by **11.95%, 14.99%, 15.43, and 15.62%** compared with that of triangular fins. The improved heating efficiency here means a relative difference in melting time between both cases at the same $\theta$.

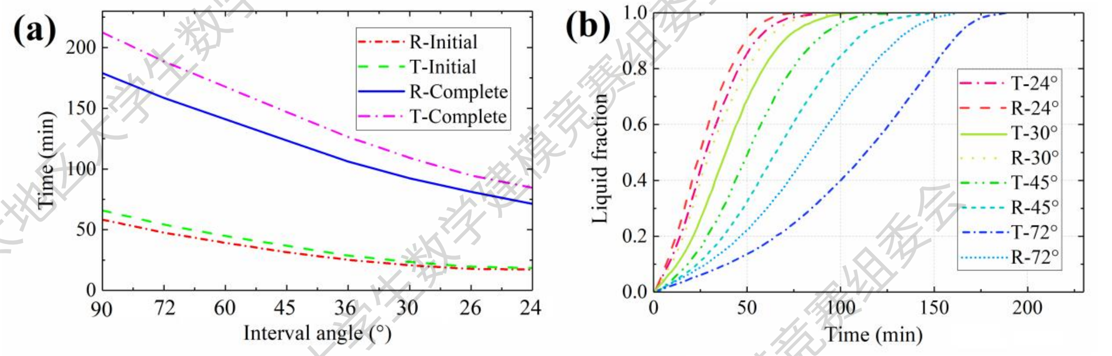



Fig. 11. Comparison between rectangular fins and triangular fins for various $\theta$: (a) phase change time, (b) liquid fraction over time

(Note: T means triangular fins, and R means rectangular fins)

## 5.2 Effect of the fin length

Four fin lengths ($L$) changing from 24mm to 6mm were studied. Fig. 12 depicts the cross-section under various $L$. The $\theta$ was 30° and and the $W$ was 6mm.

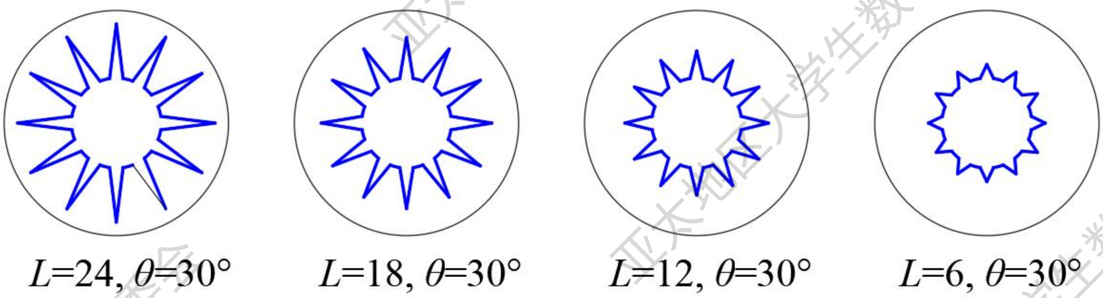



Fig. 12. Cross-section under various $L$

### 1) Influence on heat transfer

Fig. 13 shows the temperature distribution under various $L$ of triangular fins. The melting time decreases by about 70.4% when the $L$ changes from 12mm to 24mm. This is because, at the early stage of melting, the fin tip of the longer fin first contacts the PCM far away. This leads to the promotion of natural convection in the periphery of the tank. Therefore, longer $L$ can improve the heat transfer process obviously.

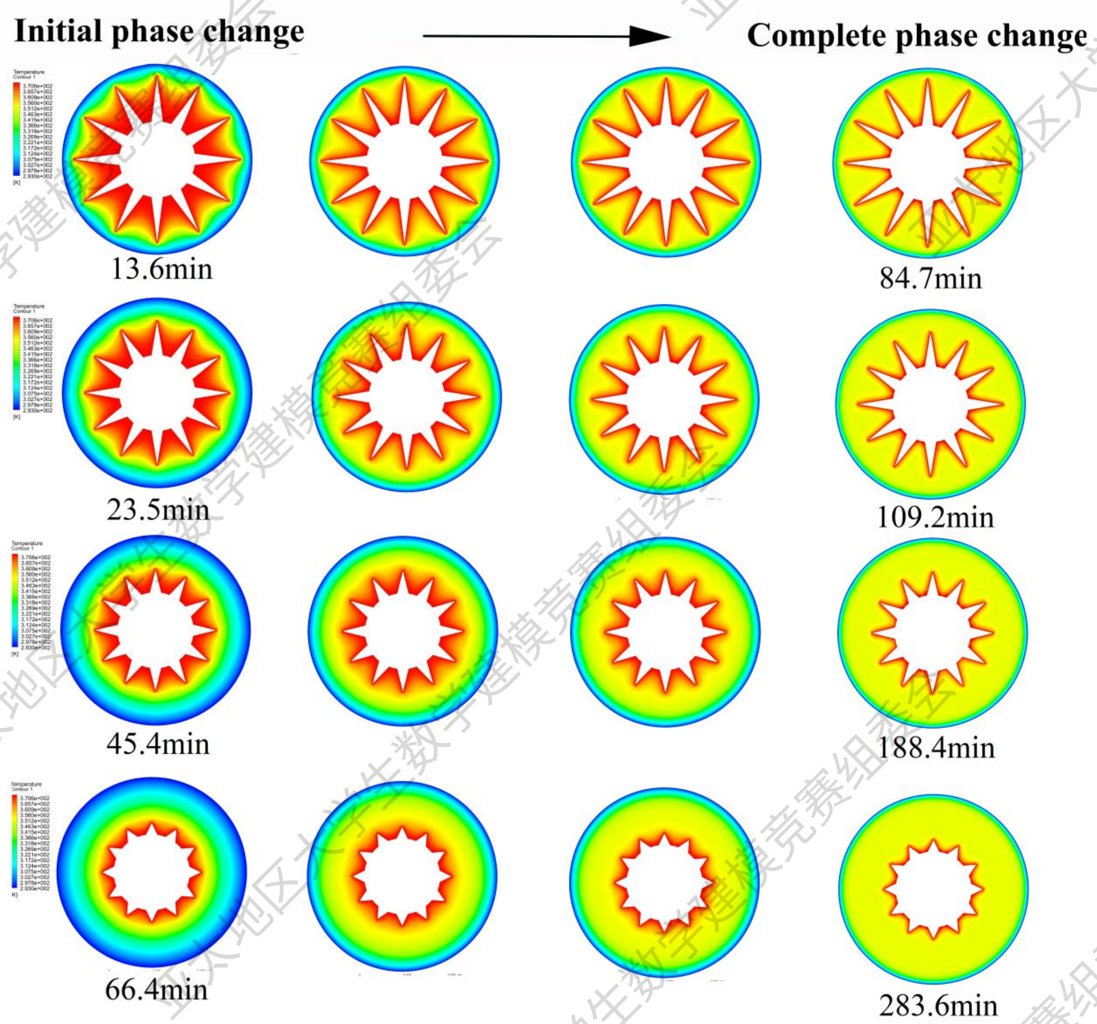



Fig. 13. Temperature distribution under different $L$

**2) Comparison and analysis**

Fig. 14(a) shows the comparison of phase change time between different $L$. The heating efficiency **significantly improves** as the increase of $L$. **For initial phase change**, $L$ =24mm can promote **79.52%, 70.04%, and 42.17%** heating rate compared with $L$ =6mm, 12mm, and 18mm, respectively. **For complete phase change**, the melting time of $L$ =24mm can save **70.13%, 55.06%, and 22.43%** compared with $L$=6mm, 12mm, and 18mm, respectively.

Fig. 14(b) shows the liquid fraction curves for different $L$ during melting. It still can be seen that long fins are more efficient than short fins.

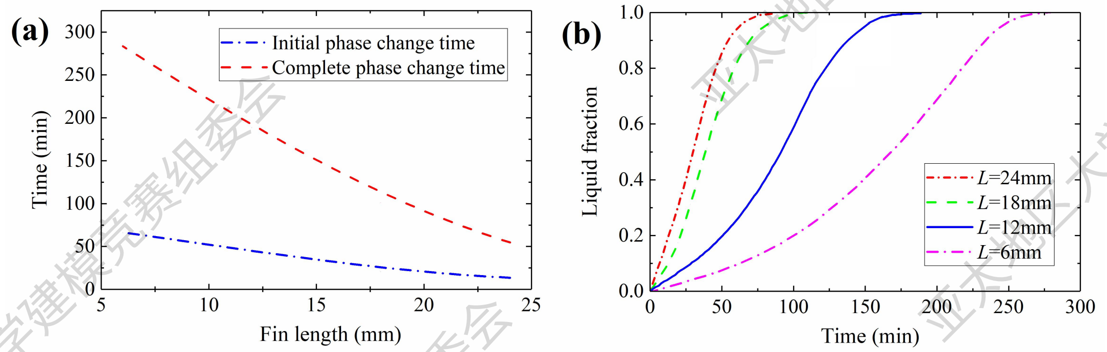



Fig. 14. Comparison between various $L$: (a) phase change time, (b) liquid fraction over time

### 5.3 Effect of the fin width

Four fin widths ($W$) changing from 10mm to 4mm were studied. Fig. 15 depicts the cross-section under various $W$. The $\theta$ was 30°and the $L$ was 18mm.

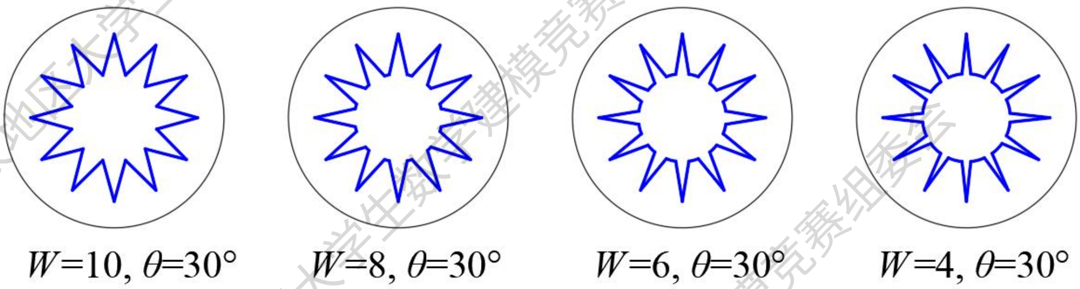



Fig. 15. Cross-section under various $W$

**1) Influence on heat transfer**

Fig. 16 shows the temperature distribution under various $W$ of triangular fins. The melting time decreases by about 35.9% when the $W$ changes from 4mm to 10mm. Therefore, larger $W$ can slightly improve the heat transfer process.

**2) Comparison and analysis**

Fig. 17(a) shows the comparison of phase change time between different $W$. The heating efficiency **slightly improves** with the increase of $W$. **For initial phase change**, $W$=10mm can promote **7.6%, 22.55%, and 35.91%** heating rate compared with $W$=8mm, 6mm, and 4mm, respectively. **For complete phase change**, the melting time of $W$=10mm can save **11.13%, 21.79%, and 30.52%** compared with $W$=8mm, 6mm, and 4mm, respectively.

Fig. 17(b) shows the liquid fraction curves for different $W$ during melting. It still can be seen that wide fins are more efficient than narrow fins.

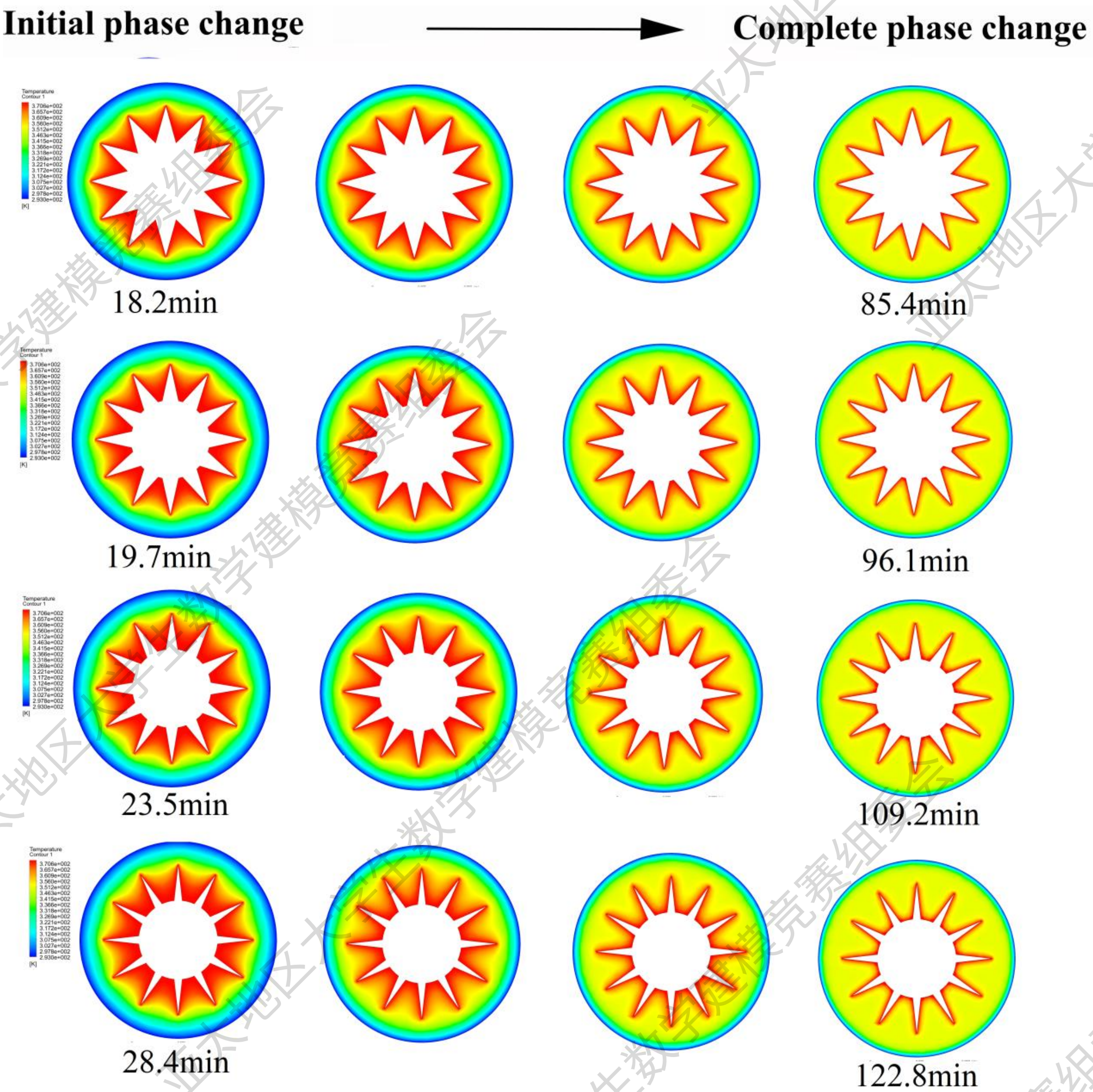



Fig. 16. Temperature distribution under different $W$

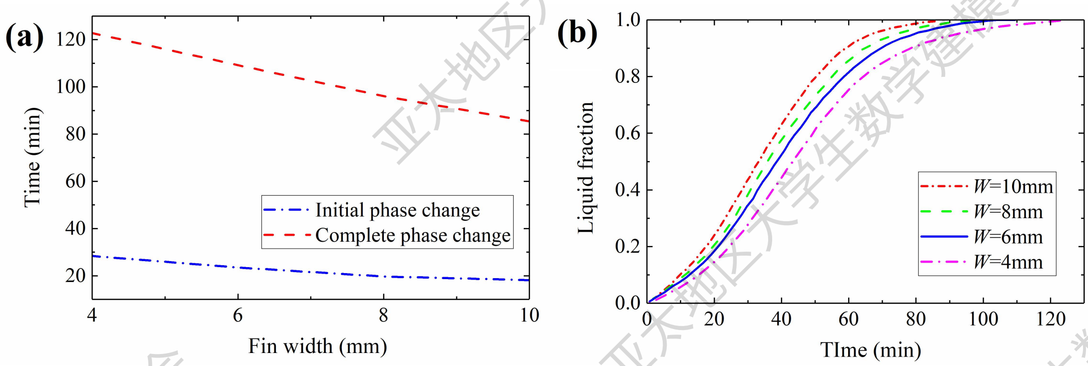



Fig. 17. Comparison between various $W$: (a) phase change time, (b) liquid fraction over time

# 6. Modeling and solution of Problem 3

## 6.1 Problem analysis

The optimal spatial distribution of fins is required to achieve the best heat transfer capacity. The optimization aim is to maximize the heat transfer efficiency in the heat storage tank.

Based on the conclusions of questions 1 and 2, we have known that the optimal geometrical shape is closely related to the length $L$, width $W$, and number $n$ of fins. For branching structure design, topology optimization and fractal design can be used to obtain the optimal spatial distribution.

**1) Design by topology optimization**

Topology optimization is a shape optimization method that uses algorithms to optimize material layout within a limited space. It maximizes the performance and efficiency of structures by removing redundant material and reaching the optimal design goal. The principle of topology optimization is shown in Fig. 18(a).

**2) Design by fractal optimization**

Fractal geometry is a geometric shape containing detailed structure at arbitrary scales, usually having a fractal dimension strictly exceeding the topological dimension. Fractal can generate the optimal fractal shape under the constraints of the objective function. Fig. 18(b) shows the principle of fractal optimization.

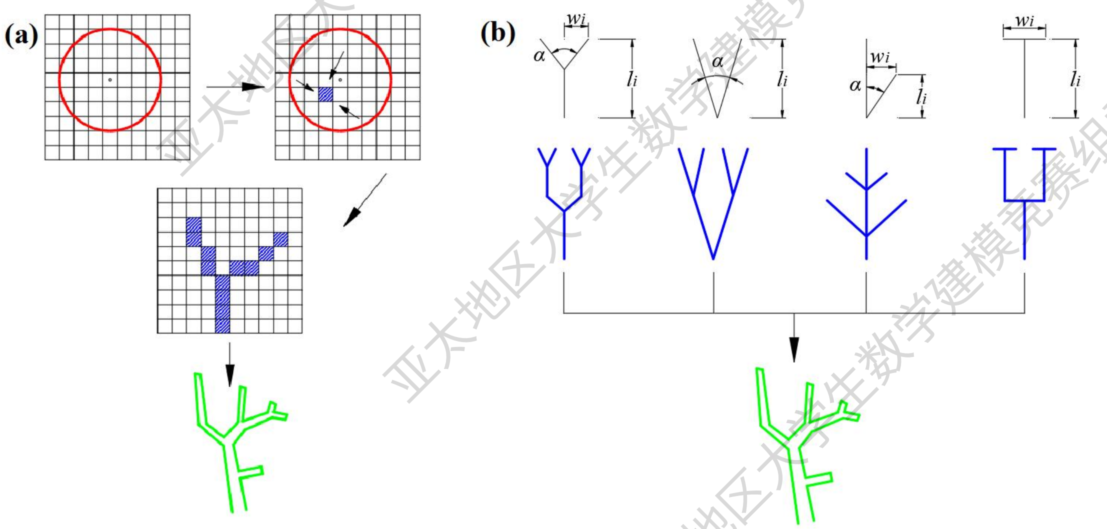



Fig. 18. Principle diagram of fin optimization: (a) topology optimization, (b) fractal design

## 6.2 Topology optimization

### 6.2.1 Variable density method

To find the optimal fin shape, the variable density method (VDM)[6] is used to optimize the spatial distribution of fins. The VDM introduces material with variable imaginary density and corresponding physical properties, to transform structure optimization problems into material layout problems.

The interpolation relationships of each variable between original and imaginary are redefined by solid isotropic material with penalization (SIMP) [6], which can be written as:

$$k(x) = (k_{HCM} - k_{PCM})(\rho_e)^{\eta} + k_{PCM} \tag{19}$$

$$\rho(x) = (\rho_{HCM} - \rho_{PCM})\rho_e + \rho_{PCM} \tag{20}$$

$$\lambda(x) = (\lambda_{HCM} - \lambda_{PCM})\rho_e + \lambda_{PCM} \tag{21}$$

$$L(x) = (1 - \rho_e)L \tag{22}$$

$$\theta(x) = (\theta_{HCM} - \theta_{PCM})\rho_e + \theta_{PCM} \tag{23}$$

where $k(x)$ is the element stiffness, $\rho(x)$, $\lambda(x)$ $L(x)$, and $\theta(x)$ are pseudothermal density, conductivity, fin length, and interval angle respectively, $\rho_e$ is the pseudo density, $\rho_e \in [0,1]$, $\eta$ is penalty factor, $\eta \in [0,1]$.

The solution steps of topology optimization based on VDM are expressed as:

$$\begin{cases} \text{find } \boldsymbol{x} = (x_1, x_2, \cdots, x_n) \\ \text{minimum } \mathbf{T} = \dfrac{1}{n}\displaystyle\sum_{i=1}^{n} \mathbf{T}_{\boldsymbol{i}} \\ \qquad \mathbf{KT} = \mathbf{Q} \\ \text{constraint } \quad \dfrac{1}{V}\displaystyle\sum_{i=1}^{n} V_i \rho_i \le \varphi \\ \qquad \rho_{min} \le \rho_i \le 1 \end{cases} \tag{24}$$

where **T** is the temperature array, **K** is the stiffness matrix, **Q** is the heat load matrix, $n$ is the number of elements, $V_i$ is the volume of element $i$, $\varphi$ is the maximum volume fraction, and $\rho_{\min}$ is the minimum pseudo density which equals $10^{-9}$ to avoid the occurrence of singular matric.

The problem from Eq. (19) to Eq. (24) is an approximate mathematical programming problem of continuous variables, whose Lagrangian function is[7]:

$$L = Q + \beta_1(V - f_v V_0) + \beta_2^T(KT - Q) + \sum_{i=1}^{n} \beta_3^i\,(\underline{x}_i - x_i) + \sum_{i=1}^{n} \beta_4^i\,(x_i - \overline{x}_i) \tag{25}$$

According to the Kuhn-Tucker rule[7], at the optimum point there is:

$$\frac{\rho(x)^{\eta} T_i k_0 T_i}{V_i} = \beta_1 \tag{26}$$

### 6.2.2 Optimization result

The optimization goal was the maximum average temperature in the PCM region. The finite element software ANSYS was still used in the optimization process for the steady-state solution. The method of moving asymptotes (MMA)[8] was used for calculation. Optimized tolerance was $10^{-3}$. The initial value of the element pseudo density $\rho_i$ was 0.5, and the maximum volume fraction $\varphi$ was set to 0.05. To get explicit topology results, the penalty factor $\eta$ is usually set to 3.

The maximum average temperature was 362.46K after 500 iterations in MATLAB. The fin distribution obtained by topology optimization is shown in Fig. 19(a).

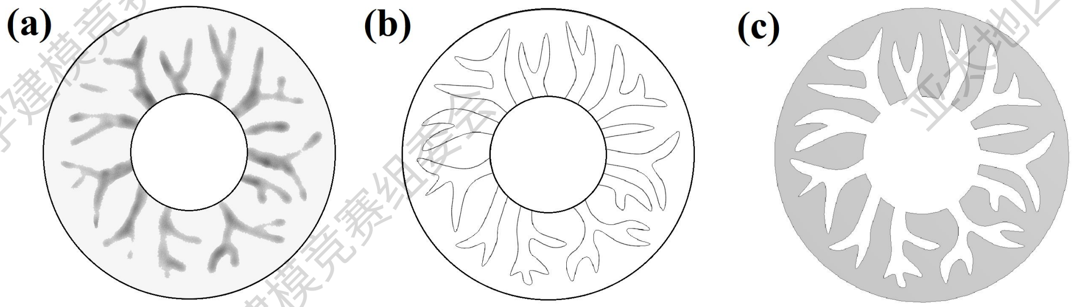



Fig. 19. Topological shape of fin distribution: (a) topology optimization, (b) image reconstruction, (c) final fin distribution used in ANSYS

### 6.2.3 Image reconstruction

Although SIMP can effectively reduce the intermediate value, the junctions between fins and PCMs exist due to the continuity of design variables. Therefore, the fuzzy boundary needs to be filtered out to make the topological graph sharper. The reconstruction function is chosen as[9]:

$$x_p = \frac{\tanh(\beta\mu) + \tanh(\beta(x_f - \mu))}{\tanh(\beta\mu) + \tanh(\beta(1-\mu))} \tag{27}$$

where $x_p$ is the mapped variable, $x_f$ is the initial variable, $\mu$ is the mapping threshold which is taken as 0.5, $\beta$ is the control parameter of mapping steepness.

The shape image of optimal fins can be obtained by drawing the node coordinates. For the convenience of numerical simulation in ANSYS, the optimized shape is converted into an 'Auto CAD' image for further processing. The reconstruction result of the fin distribution is shown in Fig. 19(b). The fin distribution used in ANSYS is depicted in Fig. 19(c).

### 6.2.4 Heat transfer process

Fig. 20 shows the temperature distribution of the topological optimization. Fig. 21 shows the phase change time and liquid fraction, respectively. Compared with rectangular fins and triangular fins, the heating efficiency of topology optimization fins significantly improves. **For initial phase change**, topology optimization fins can promote **18.84% and 28.17%** heating rate compared with rectangular fins and triangular fins. **For complete phase change**, the melting time of topology optimization fins can save **41.81% and 50.82%** compared with rectangular fins and triangular fins, respectively.

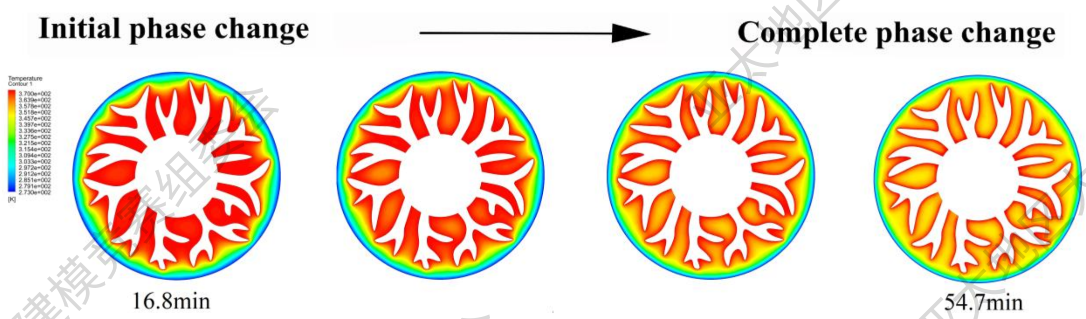



Fig. 20. Temperature distribution of topological optimization shape

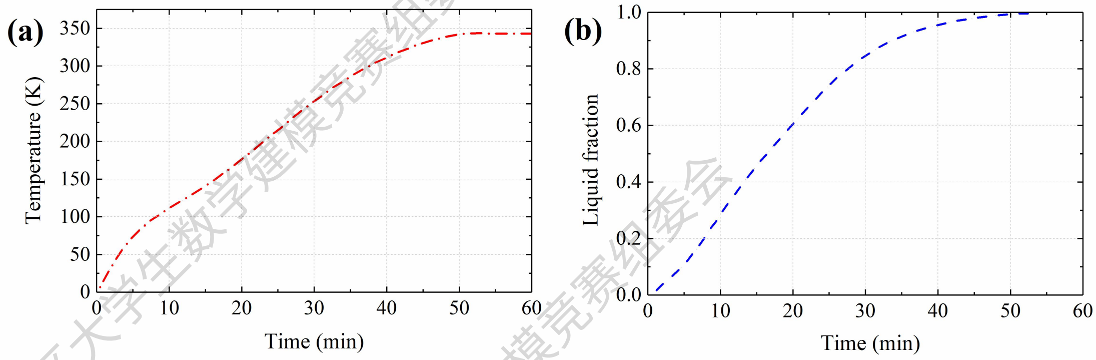



Fig. 21. Temperature variation and liquid fraction

### 6.3 Fractal optimization

As can be seen that the optimized fin structure obtained by topology optimization is irregular, which may cause difficulties in the manufacture and application of fin components. Therefore, A regular fin distribution is proposed optimally by fractal optimization in this section.

#### 6.3.1 Murray's law and branch generation

The fractal fins have statistically self-similar and multi-scale properties, which can be considered as a mimicking of a fern leaf. The optimal transportation of leaf is derived by a relationship known as Murray's law[10]:

$$r^3 = r_1^3 + r_2^3 + r_3^3 + \cdots + r_n^3 \tag{28}$$

where is the radius of the parent tube, $r_1$ to $r_n$ are the radii of the branching tube. Based on the coupling of Murray's law, the generalized Murray's law can be obtained as[11]:

$$r^\gamma = \frac{1}{1-\zeta}\sum_{i}^{n} r_i^\gamma \tag{29}$$

where $\zeta$ is the heat variation ratio, $\gamma$ is determined by the mode of transport, and $n$ is the number of child tubes.

The fern-fractal fins were designed by the generalized Murray's law. The exponent $\gamma$ was taken 2 to achieve better thermal performance for the heat conduction [12]. The construction method of the fractal fin was introduced as follows:

**Step 1:** the parent branch grows two children branches. The length and width of the parent branch were $L_0$ and $W_0$, respectively. The length and width of the first level of children's branches were $L_1$ and $W_1$, and the second level were $L_2$ and $W_2$. The terminals of the children branches were close to the outer wall of the tank.

**Step 2:** The lengths of any two successive branches (i.e., the parent branch and the middle child branch) meet the following relationship, which is written as

$$L_i = \alpha L_{i-1} \tag{30}$$

where $\alpha$ is the length ratio that reflects the length growth law.

**Step 3:** The correlation between the widths of the parent and children branches can be calculated by

$$W_i^2 = \frac{1}{3} W_{i-1}^2 \tag{31}$$

**Step 4:** Repeat the above steps 1 to 3 until the desired branching level. The number of branches per level was optimized by maximum average temperature as well. Then the fern-fractal fins are constructed.

### 6.3.2 Optimization result

The fin distribution obtained by Fractal design is shown in Fig. 22(a). The fin distribution used in ANSYS is depicted in Fig. 22(b).

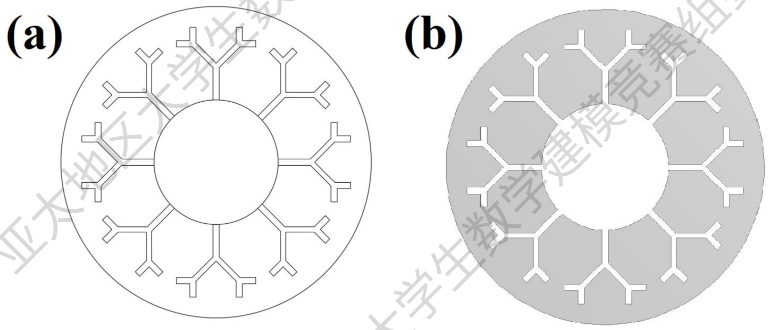



Fig. 22. Fractal shape of fin distribution: (a) fractal optimization, (b) finial fin distribution used in ANSYS

### 6.3.3 Heat transfer process

Fig. 23 shows the temperature distribution of the fractal optimization. Fig. 24 shows the phase change time and liquid fraction, respectively. Compared with rectangular fins and triangular fins, the heating efficiency of fractal optimization fins significantly improves. **For initial phase change**, fractal optimization fins can promote **14.01% and 24.25%** heating rate compared with rectangular fins and triangular fins, respectively. **For complete phase change**, the melting time of topology optimization fins can save **32.57% and 42.94%** compared with rectangular fins and triangular fins, respectively.

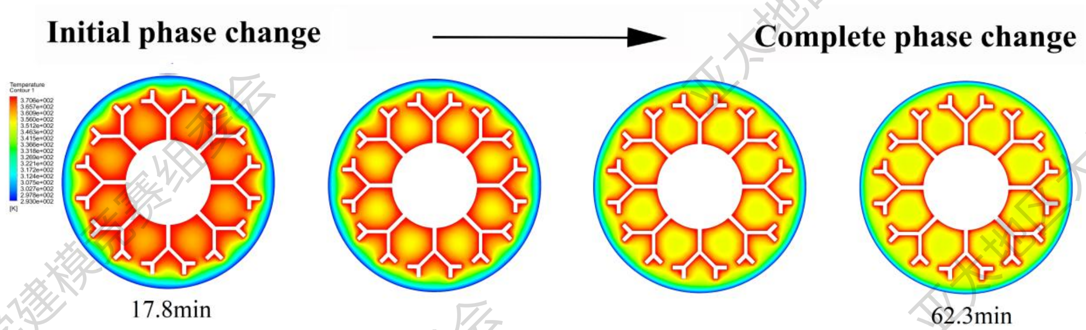



Fig. 23. Temperature distribution of fractal optimization shape

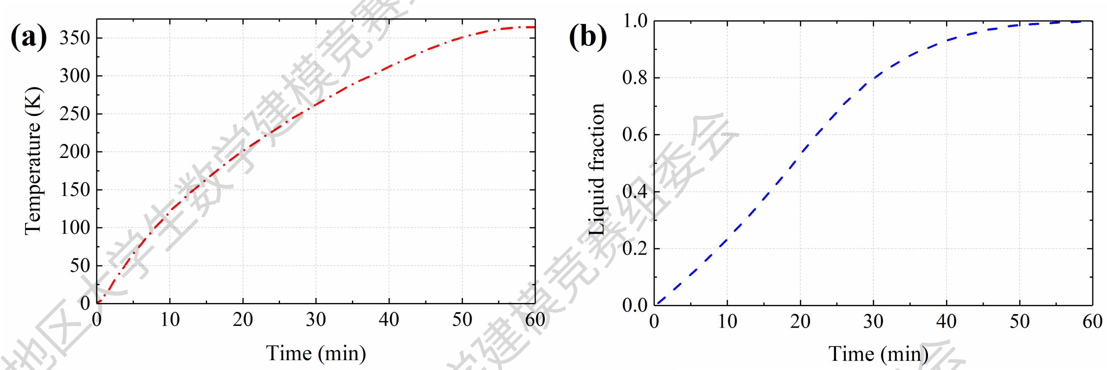



Fig. 24. Temperature variation and liquid fraction

## 6.4 Comparison of two models

According to the phase change time and liquid fraction, topology optimization is more efficient than fractal optimization. For initial and complete phase change time, **topology optimization increases by 5.61% and 13.8% heating rate compared with fractal optimization.**

# 7. Recommendation letter

Dear company leaders,

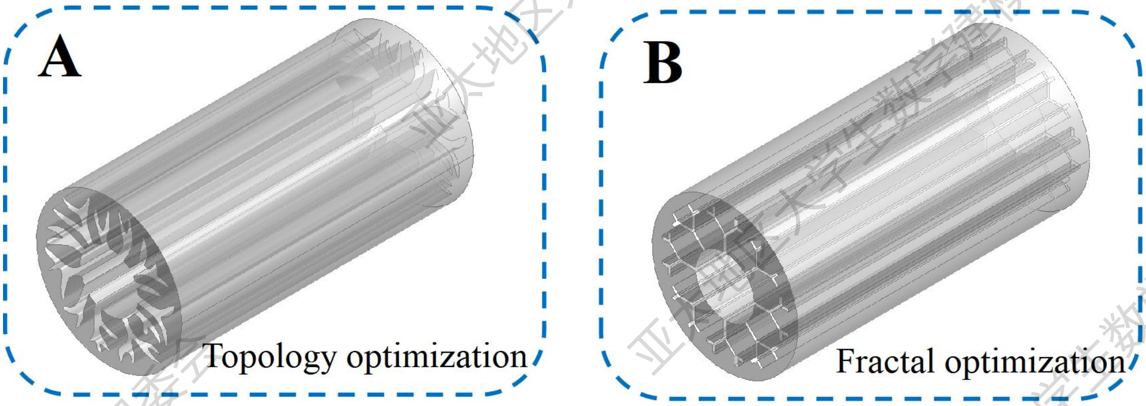
We are writing this letter to provide you with some suggestions to optimize the heat transfer fins in the heat storage tank. To achieve the optimal fin configurations, We have devised two optimization models to obtain the optimal fin configurations. The design suggestions are summarized based on the model simulation and structural analysis.

First, parametric studies of fins were conducted to preliminarily investigate the effects of shape, geometry, and spatial distribution on heat transfer efficiency. Generally speaking, fins configurations with smaller interval angles and larger lengths and widths are more efficient. The priority order of fin parameters is **fin length > interval angle > fin width**. But the thermal efficiency will weaken when the fin configurations exceed a certain limited value. Based on our research, **the recommended ranges of design parameters** are as follows:

| Parameter | Interval angle (°) | Length (mm) | Width (mm) |
|---|---|---|---|
| Recommended range | 24~45 | 18~24 | 6~10 |

In addition, topology optimization was conducted to generate an arborescent fin distribution as shown in Fig. A. However, the optimized fin structure obtained by topology optimization is **irregular** most of the time, which may cause difficulties in the manufacture and application of fin components. Therefore, I suggest that this topology optimization structure can be adopted in **PCMs with very low thermal conductivity**.

Finally, fractal optimization was performed to obtain a **regular** fin distribution as shown in Fig. B. This method provides more possibilities for engineering applications. But it should be pointed out that the hearing efficiency of fractal optimization is less efficient than that of topology optimization. Therefore, this fractal optimization can be used in **general heat storage tanks** and **PCMs with low thermal conductivity**.




We hope that the proposed models, results, and suggestions could provide you with valuable information.

Yours sincerely,

Team Dapmcm2201058

# 8. Model evaluation and promotion

### 7.1 Evaluation

The sensitivity of exponent $\gamma$ in the fractal model was tested by changing the fractal dimension and comparing the difference between different configurations. The exponent $\gamma$ was taken 2, 3, and 4, respectively, to investigate the influence of fractal dimension (i.e. bifurcation number) on the temperature variation.

Fig. 25 shows the comparisons of temperature variation under different $\gamma$. It can be seen that the temperature rise is slightly promoted with the increase of $\gamma$. This is because no significant change in the spacing between adjacent fins due to the control of fractal tree. Therefore, the fractal design model has good stability.

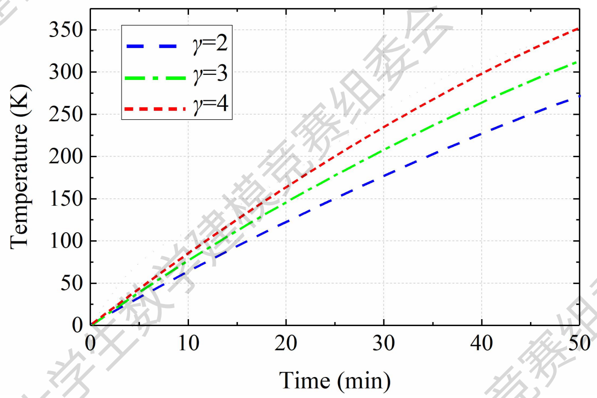



Fig. 25. Comparisons of temperature variation under different $\gamma$

### 7.2 Promotion

1) In order to obtain the real features of heat transfer, the viscous dissipation and thermal expansion should be considered in Eq. (7) and (9), because large velocity gradients may occur during the heat flow in the PCM domain.

2) Considering the actual structure, three-dimensional topology optimization should be conducted to obtain a tridimensional structure. The fins can be designed along the longitudinal direction or layered layout.

# 9. Strength and weakness

### 7.1 Strength

1) When predicting the optimal fin distribution, the fractal optimization model has **better model stability** compared with the topology optimization model. However, it also should be pointed out that the fin distribution obtained by topology optimization us more efficient for the heat transfer.

2) The topology optimization model can find the optimal fin distribution **considering various balance factors** in optimization objectives.

3) The proposed topology and fractal optimization models can **automatically**

**search** the optimal fin distribution under the constraint of objective functions.

### 7.2 Weakness

1) Most of the time, the optimized shape obtained by topology optimization is irregular, which causes difficulties in the manufacture and application of fin components.

2) Objective function plays a critical role in the final optimization results. When selecting different optimization objectives, the final optimization results may be different. Therefore, objective function could be diverse in solving the optimal fin distribution.

# Reference


[1] Voller VR, Brent AD, Prakash C. Modelling the mushy region in a binary alloy [J]. Applied Mathematical Modelling, 1990, 14(6):320-326.

[2] Cheng Y, Xuan ZB, Xi CB, et al. Melting performance enhancement of a latent heat storage unit using gradient fins [J]. International Journal of Heat and Mass Transfer, 2022, 150:11930.

[3] Al-Abidi AA, Mat S, Sopian K, et al. Internal and external fin heat transfer enhancement technique for latent heat thermal energy storage in triplex tube heat exchangers [J]. Applied Thermal Engineering, 2013, 53(1):147-156.

[4] Seddegh S, Joybari MM, Wang X, et al. Experimental and numerical characterization of natural convection in a vertical shell-and-tube latent thermal energy storage system [J]. Sustainable Cities and Society, 2017, 35:13-24.

[5] Mao Q, Li Y, Li G, et al. Study on the influence of tank structure and fin configuration on heat transfer performance of phase change thermal storage system [J]. Energy, 2021, 235:121382.

[6] Bendse MP, Sigmund O. Material interpolation schemes in topology optimization [J]. Archive of Applied Mechanics, 1999, 36: 635-654.

[7] Geng DC. Basis of engineering structure optimization design [M]. Dalian University of Technology Press, 2012.

[8] Krister D. The method of moving asymptotes - a new method for structural optimization [J]. International Journal of Numerical Methods Engineering, 1987, 24:359-373.

[9] Zhao CY, Wu ZG. Heat transfer enhancement of high temperature thermal energy storage using metal foams and expanded graphite [J]. Solar Energy Materials and Solar Cells, 2011, 95(2): 636-43.

[10] Mcculloh KA, Sperry JS, Adler FR. Water transport in plants obeys Murray's law [J]. Nature, 2003, 421:939-942.

[11] Zhou Y, Kassab GS Molloi S. On the design of the coronary arterial tree: a generalization of Murray's law [J]. Physics in medicine and biology, 1999, 44(12):2929-2945.

[12] Xu P, Sasmito AP, Yu B, et al. Transport phenomena and properties in treelike networks [J]. Applied Mechanics Reviews, 2016, 68:1-9.